



EUNOIA JUNIOR COLLEGE
JC2 Preliminary Examination 2021
General Certificate of Education Advanced Level
Higher 1

CANDIDATE
NAME

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CIVICS
GROUP

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REGISTRATION
NUMBER

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CHEMISTRY

Paper 2 Structured Questions

8873/02

16 September 2021

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Do not use paper clips, highlighters, glue or correction fluid.

Write your name, civics group, registration number on all the work you hand in.

Write in **dark blue or black pen**.

You may use an HB pencil for any diagrams or graphs.

Section A

Answer **all** questions.

Section B

Answer **one** question.

The number of marks is given in brackets [] at the end of each question or part question.

A Data Booklet is provided.

The use of an approved scientific calculator is expected, where appropriate.

For Examiner's Use	
Section A	
1	/ 14
2	/ 8
3	/ 15
4	/ 23
Section B	
5 / 6	/ 20
Total	/ 80

This document consists of **23** printed pages and **1** blank page.

Section A

Answer all questions in this section.

- 1 The fourth ionisation energies of the elements **A** to **G**, of proton number ranging from 6 to 20, are shown in Fig. 1.1.

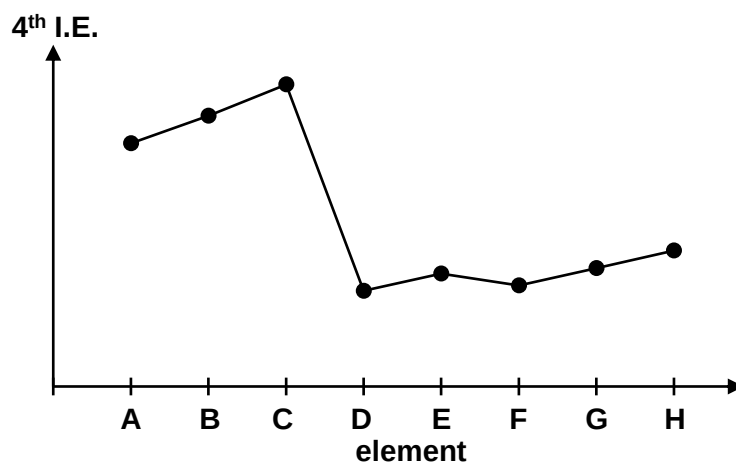


Fig. 1.1

- (a) (i) Explain why element **F** is sulfur.

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..... [2]

- (ii) Write the electronic configuration of F^{3+} and E^{3+} . Hence, explain why element **F** has a lower 4th ionisation energy than element **E**.

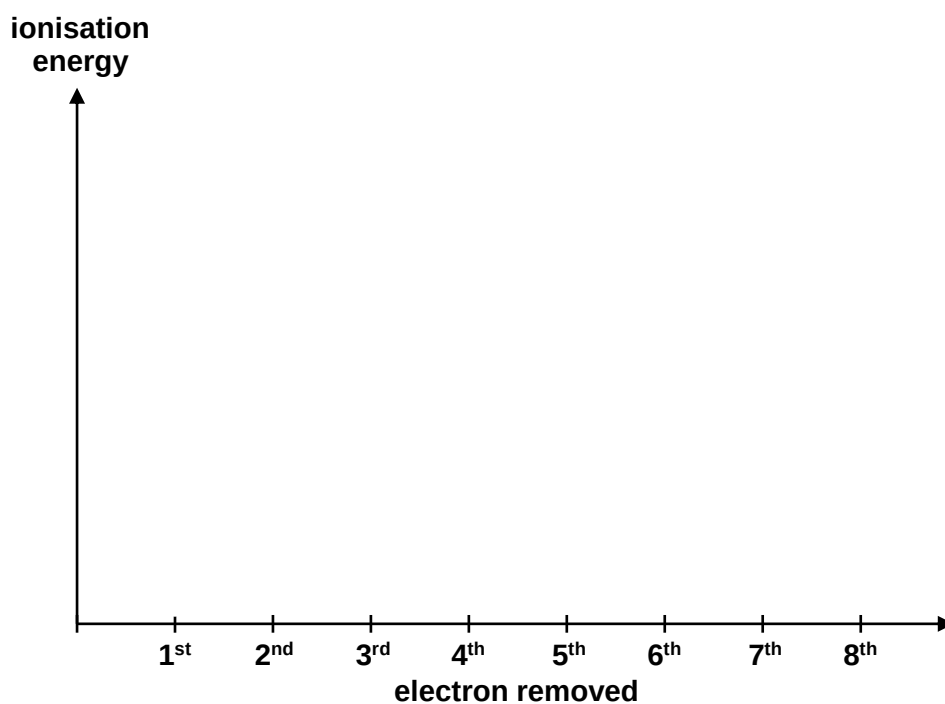
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..... [2]

(iii) Sketch the first 8 successive ionisation energies of element **F** in the axes below.



[1]

(b) Oxides of Period 3 elements exhibit acid/base behaviour. With the aid of suitable equations, describe the nature of sodium oxide, aluminium oxide and phosphorous pentoxide when reacted with hydrochloric acid and/or sodium hydroxide.

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.....[3]

- (c)** Magnesium chloride, silicon tetrachloride and phosphorus pentachloride are some of the chlorides of Period 3 elements. With reference to structure and bonding, rank the melting points of these three chlorides in decreasing order. Explain your answer.

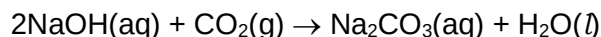
.....[3

- (d)** When aluminium chloride is added to water with universal indicator, the solution turned orange. A similar observation was made when iron(III) chloride was added to water. Using appropriate values from the *Data Booklet*, suggest a reason and write the equation(s) of the reaction of iron(III) chloride with water.

.....[3]

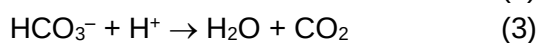
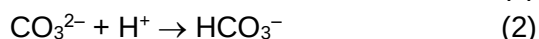
[Total: 14]

- 2 4.50 g of a monobasic organic acid, $C_xH_{2x+1}CO_2H$, is burnt completely in excess of oxygen. The carbon dioxide gas evolved is completely absorbed into 1 dm³ of sodium hydroxide solution:



To determine the concentration of sodium hydroxide used, the resultant solution was titrated against hydrochloric acid solution.

During the titration, the reaction between sodium hydroxide and hydrochloric acid occurs in a single step, while that between sodium carbonate and hydrochloric acid occurs in two steps with hydrogencarbonate ion as the intermediate product.



With phenolphthalein as indicator, the end-point is seen at the completion of reactions (1) and (2). With methyl orange as indicator, the end-point is seen at the completion of all three reactions.

- (a) Suggest a reason why excess sodium hydroxide is used to absorb carbon dioxide gas.

.....[1]

- (b) From the 1 dm³ resultant solution, 10 cm³ samples were titrated against 0.250 mol dm⁻³ hydrochloric acid. When one sample was titrated in the presence of phenolphthalein, 16.00 cm³ of hydrochloric acid was required to reach the end-point. When a second sample was titrated in the presence of methyl orange, 24.20 cm³ of hydrochloric acid was required to reach the end-point.

- (i) Calculate the amount of hydrochloric acid that has reacted in both samples. Hence, show that in a 10 cm³ sample, the amount of hydrogencarbonate ion present is 0.00205 mol.

[2]

- (ii) Using the value given in (b)(i), calculate the respective amount of carbonate ions and hydroxide ions present in the resultant solution.

[2]

- (iii) Hence, calculate the concentration of sodium hydroxide used to absorb carbon dioxide gas.

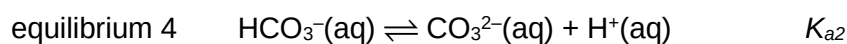
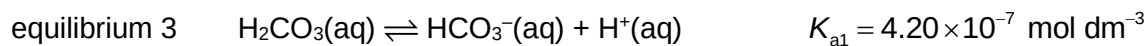
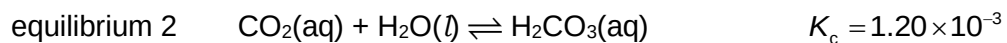
[1]

- (iv) Using the value given in (b)(i), deduce the value of x in the organic acid, $C_xH_{2x+1}COOH$.

[2]

[Total: 8]

- 3 When carbon dioxide from the atmosphere dissolve in the ocean, it forms carbonic acid, H_2CO_3 , which in itself is unstable. This further dissociates to form hydrogencarbonate and carbonate ions.



where K_{a2} is very small.

- (a) With reference to structure and bonding, suggest why $\text{CO}_2(\text{g})$ dissolves better in water than $\text{O}_2(\text{g})$ and $\text{N}_2(\text{g})$.

.....

 [2]

- (b) (i) Write the K_{a1} expression for equilibrium 3.

[1]

- (ii) A sample of ocean water has a pH of 8.1. What is the concentration of H^+ ions in ocean water?

[1]

- (iii) Assuming the concentration of H^+ in ocean water is mainly from equilibrium 3, use your answer in (b)(ii) to calculate the equilibrium concentration of $\text{H}_2\text{CO}_3(\text{aq})$ in equilibrium 3.

[2]

(iv) Hence, calculate the equilibrium concentration of dissolved CO_2 in equilibrium 2.

[1]

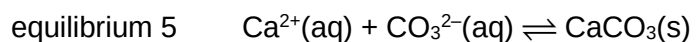
- (c) Since the Industrial Revolution, human activity has increased the level of atmospheric CO_2 , which in turn lowered the pH of the ocean. Using equilibrium 1 to 4, explain this phenomenon.

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.....[4]

- (d) The $\text{HCO}_3^- / \text{CO}_3^{2-}$ system acts as a buffer for the ocean. State the equation showing how the buffer responds to the lowering of pH in the ocean.

.....[1]

Marine organisms such as mussels and corals use the abundant calcium ions present in the ocean to form CaCO_3 shells and exoskeletons.



- (e) (i)** Using your answer in **(d)** and equilibrium 5, suggest the impact of sustained increased level of atmospheric CO_2 on marine organisms such as corals and mussels.

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[1]

- (ii)** Explain if equilibrium 5 is exothermic or endothermic, given that $\text{CaCO}_3(\text{s})$ dissolves less easily with rising temperature.

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[2]

[Total: 15]

- 4 Poly(ethene) is the simplest addition polymer containing only singly bonded carbon and hydrogen atoms. It is durable, tough and have excellent electrical insulating properties.

Ethene, CH_2CH_2 , forms poly(ethene) of different molecular structures by varying temperature, pressure and catalyst used.

Table 4.1 shows some data for poly(ethene) formed under different conditions.

Table 4.1

	density / g cm^{-3}	tensile strength / MPa	elongation at fracture / %	uses
A	0.92	15	600	plastic wrap
B	0.97	29	350	milk bottles
cross-linked poly(ethene) (CLPE)	0.93	18	270	bottle caps

Tensile strength is the amount of load or stress that can be handled by a material before it stretches and breaks.

Elongation at fracture is the ratio between changed length and initial length after breakage of the test specimen.

- (a) (i) Explain what is meant by the term *polymer*.

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 [1]

- (ii) Briefly describe in terms of bonding, what happens during the addition polymerisation of ethene.

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 [1]

- (iii) Explain the change in bond angle about carbon atom from ethene to poly(ethene), using VSEPR theory.

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 [2]

- (iv) The formation of poly(ethene) is catalysed by solid chromium oxide, a heterogeneous catalyst. Outline the mode of action for this type of catalysis.

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.....[3]

- (b) Suggest if any of the polymers in Table 4.1 will sink in water.

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.....[1]

- (c) Using information from Table 4.1 and your knowledge of the structure and bonding in polymers, answer the following questions:

- (i) **A** and **B** could be either LDPE or HDPE. Deduce, with reasoning, the identity of **A** and **B**.

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.....[4]

- (ii) With reference to your answer in (c)(i), state a relevant physical property that allow **A** and **B** to be used according to table 4.1.

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.....[2]

- (d) Among the three types of poly(ethene), CLPE is unsuitable for recycling or remoulding. Suggest, with reasoning whether CLPE is a thermoplastic or thermoset polymer.

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.....[2]

- (e) The durable properties of poly(ethene) also posed environmental issues.

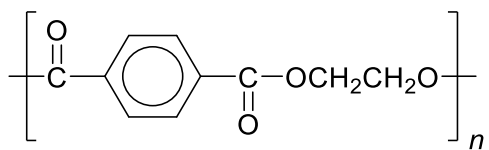
- (i) Suggest why poly(ethene) is non-biodegradable.

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.....[1]

- (ii) Carbonyl groups (C=O) are incorporated into poly(ethene) to allow it to become photodegradable by absorbing UV light. Suggest one situation where the use of photodegradable polymer is unsuitable.

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.....[1]

- (f) Poly(ethylene terephthalate) are also used to make bottles.



poly(ethylene terephthalate) or PET

PET bottles are not suitable to store strongly alkaline solutions. State the reaction that may occur between the PET bottle and the alkaline solution and draw the structure of products formed.

.....[2]

- (g) Ethyne, the simplest alkyne, has the structure $\text{HC}\equiv\text{CH}$. Similar to ethene, ethyne undergoes addition polymerisation to form poly(ethyne).

The discovery and development of poly(ethyne) as an organic conducting polymer led to Nobel Prize in Chemistry in 2020. The electrical conductivity of poly(ethyne) is increased when small amount of iodine is added. In this reaction, iodine acts as an oxidising agent.

- (i) Draw three repeating units of poly(ethyne).

[1]

- (ii) With reference to your answer in (g)(i), account for the electrical conductivity of poly(ethyne) chain.

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.....[1]

- (iii) Suggest the effect of adding bromine instead of iodine to the electrical conductivity of poly(ethyne).

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.....[1]

[Total: 23]

Section B

Answer **one** of the questions in this section.

- 5 (a) Hydrazine is an inorganic compound with the chemical formula, N_2H_4 . It is a colourless flammable liquid with an ammonia-like odour.

- (i) Draw the structure of hydrazine, stating the shape and bond angle about each central atom, N.

Shape about N:

Bond angle:

[3]

- (ii) Hydrazine can be synthesised via the following reaction under aqueous conditions.



State and explain the type of reaction shown above.

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.....[2]

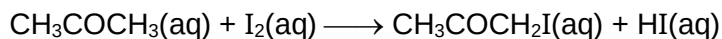
(iv) Calculate the rate constant, k , for this reaction. Include in your answer, the units of k .

.....[2]

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experiment	$[\text{ClO}^-] / \text{mol dm}^{-3}$	$[\text{NH}_3] / \text{mol dm}^{-3}$	initial rate $/ \text{mol dm}^{-3} \text{ s}^{-1}$
1	0.200	0.100	0.256
2	0.400	0.200	2.05
3	0.400	0.400	8.2

- (b) In the presence of an acid catalyst, propanone, CH_3COCH_3 , reacts with iodine according to the following equation.



In a kinetic study, $0.200 \text{ mol dm}^{-3}$ propanone, $0.0100 \text{ mol dm}^{-3}$ iodine and $0.200 \text{ mol dm}^{-3}$ acid was added to a beaker and allowed to react. At fixed time interval, 10 cm^3 samples of the reaction mixture was drawn and analysed. The graph in Fig. 5.1 was obtained.

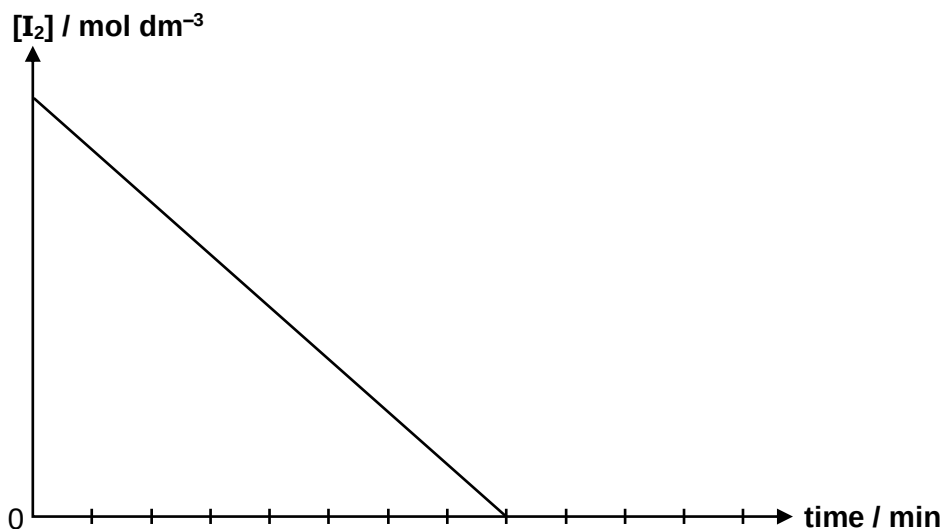


Fig. 5.1

- (i) Suggest how the analysis of the 10 cm^3 samples can be done to monitor rate of reaction.

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[1]

- (ii) Explain, with a suitable diagram, why the use of a catalyst speeds up the rate of reaction between propanone and iodine.

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..... [3]

- (iii) Suggest a reason why the order of reaction with respect to the acid catalyst cannot be deduced in this kinetic study.

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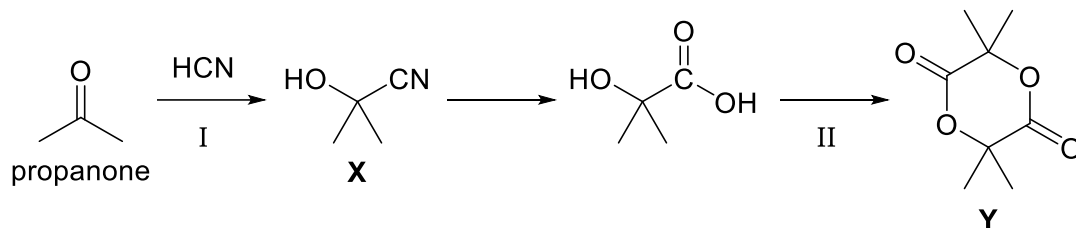
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..... [1]

- (iv) Given that the order of reaction with respect to acid is one, sketch on **Fig. 5.1**, the graph when concentration of acid is doubled while keeping all other concentrations and factors constant. [1]

- (c) Propanone can also undergo a series of reactions to form carboxylic acid as shown below.



- (i) Nitrile, $-\text{CN}$, is one of the functional groups present in compound **X**. State the other functional group present in **X**.

..... [1]

- (ii) State the type of reaction that has occurred in step I.

..... [1]

- (iii) State the reagent and condition required for step II.

..... [1]

- (iv) In addition to compound **Y**, in the presence of the reagent and condition stated in (c)(iii), compound **Z**, $\text{C}_4\text{H}_6\text{O}_2$, was formed. Draw the skeletal formula of compound **Z**.

[1]

[Total: 20]

6 Sodium is a commercially important alkali metal. As sodium is highly reactive, the elemental metal is stored under oil.

(a) Besides sodium, other group 1 metals also have to be stored carefully to prevent reacting with the atmospheric moisture and oxygen.

(i) State the type of reaction that group 1 metal undergoes with oxygen.

..... [1]

(ii) Suggest, with reasoning, the reactivity of group 1 metals with oxygen.

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..... [2]

(iii) Group 1 metals, except lithium, do not react readily with nitrogen. Using the *Data Booklet*, explain why only lithium nitride, Li_3N can be formed but sodium nitride, Na_3N cannot be formed.

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..... [2]

(iv) Group 1 metals, however, can form stable compounds with the azide ion, N_3^- .

Azide ion contain two double bonds.

Draw the dot and cross diagram of azide ion, stating its shape and bond angle.

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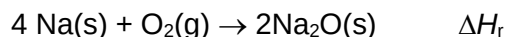
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..... [2]

- (b) Sodium burns in oxygen to form sodium oxide, Na_2O . The enthalpy change of reaction between sodium and oxygen is given below.

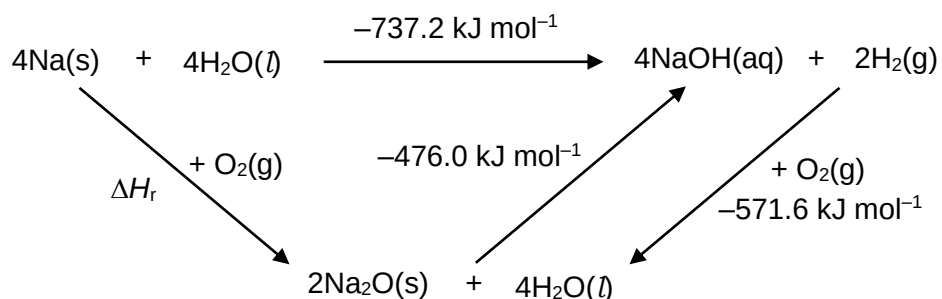


- (i) Define *Hess Law*.

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 [1]

- (ii) Apply *Hess Law* and calculate the enthalpy change of reaction, ΔH_r , using the energy cycle below.



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 [2]

- (iii) With the aid of a suitable equation, define the term *enthalpy change of combustion* of sodium.

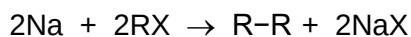
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 [2]

- (iv) Hence, determine a value for the enthalpy change of combustion of sodium using your answers to (a)(ii) and (iii).

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 [1]

- (c) In the Wurtz reaction, sodium reacts with halogenoalkane, RX, to form larger alkane:



- (i) State the type of reaction that occurred to RX, during the Wurtz reaction.

..... [1]

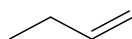
- (ii) Name the simplest straight chain alkane that can be formed the Wurtz reaction.

..... [1]

- (iii) Draw the skeletal formula of corresponding product formed when 2-chlorobutane is used in the Wurtz reaction.

[1]

- (iv) A possible side reaction is the elimination reaction of halogenoalkanes to form alkenes. One such alkene formed from reaction in (c)(iii) is given below:



Draw the other possible alkene that can be formed and suggest, with reasoning, if this other alkene exhibit *cis-trans* isomerism.

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 [2]

- (v) In another reaction, two different halogenoalkanes are reacted together with sodium.



Draw the three alkanes that can be formed.

[2]

[Total: 20]

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